# Haozhe Lei

**Ph.D. Candidate, Electrical and Computer Engineering | Expected May 2027**

RF and Multimodal Sensing | Probabilistic Localization | Robotics and Autonomous Systems

Brooklyn, NY | hl4155@nyu.edu | Homepage | Google Scholar

## SUMMARY

ECE Ph.D. candidate developing physics-informed machine learning and probabilistic spatial inference for RF sensing, localization, and autonomous systems. Leads algorithm and neural-model design, Python/PyTorch research prototypes, and evaluation across wireless digital twins and real robotic experiments. Connects uncertainty-aware RF perception with learning-based navigation and adaptive wireless decision-making.

**Research record:** 13 peer-reviewed journal/conference papers and 5 manuscripts, with first/co-first and corresponding-author leadership across core research.

## TECHNICAL SKILLS

**Programming and compute:** Python, PyTorch, MATLAB, R; Linux, shell scripting, Kubernetes
**Probabilistic inference:** Likelihood and conditional-posterior estimation, Bayesian multi-observation fusion, Monte Carlo marginalization, uncertainty calibration
**Machine learning and decisions:** U-Net, attention and Transformer models; physics-informed and neuro-symbolic reinforcement learning, PPO, recurrent actor–critic methods, bandits
**Simulation and modeling:** Sionna RT, wireless digital twins, real-to-sim calibration, measured antenna-pattern integration, TX/RX pose and mobility simulation, Wireless InSite
**Robotics and RF sensing:** TurtleBot4/ROS 2 RF–sensor synchronization; Jackal UGV FR3 payload hardware and 3D LiDAR/RGB-D sensing; SLAM-based navigation; Pi-Radio SDR channel sounding; TI radar point-cloud filtering; CARLA, SUMO

## EDUCATION

**New York University** Expected May 2027
Ph.D., Electrical and Computer Engineering
Advisor: Prof. Sundeep Rangan, Director, NYU WIRELESS | Ernst Weber Fellowship, 2023

**New York University** May 2022
M.S., Computer Engineering

**China Agricultural University** June 2019
B.E., Electrical Engineering and Automation

## RESEARCH AND ENGINEERING EXPERIENCE

**NYU WIRELESS, New York University** *Brooklyn, NY*
*Graduate Research Assistant* Sept. 2022 – Present

- **Probabilistic RF localization and efficient inference:** Lead model-based likelihood scoring and neural posterior estimation for RF source localization under sparse observations, partial maps, and propagation-model mismatch (*MC-CLE*, *LOCUS-DT*, *MAPLE-RF*). Design U-Net/attention spatial estimators, Bayesian evidence fusion, and uncertainty calibration. MAPLE-RF achieved a **218× fresh-query speedup** over full-grid general-purpose ray tracing in a single-query H100 simulation benchmark, excluding shared RF estimation.
- **Robotic sensing and real-to-sim calibration:** Designed *MAGNETAR*'s joint RF transmitter-position/heading posterior; improved posterior quality over simulation-only training using measurement-aligned simulation and limited real-data adaptation. Coordinate *MobiFR3* architecture and experiments, synchronizing Pi-Radio RF with TurtleBot4/ROS 2, LiDAR, RGB, and odometry. The latest Jackal configuration adds an FR3 payload and 3D LiDAR/RGB-D sensing. Mentor students in RF operation and data collection; validate algorithms on held-out physical measurements.

- **Learning-based control and autonomous navigation:** Lead multi-band digital-twin and learning research (*MCMB-HDT*), designing Transformer rate prediction from sparse, asynchronous histories and recurrent PPO policies for antenna selection under partial observability and sensing costs. Co-developed *PIRL* and symbolic program priors for zero-shot navigation in unseen indoor layouts, using physics-informed rewards and ray-traced propagation cues.
- **Research leadership and spatial intelligence:** Mentor Ph.D. and M.S. researchers across wireless learning, RF–robot experiments, and object-centric RGB-D scene memory. Guide method development, experimental evaluation, and embodied-agent pipelines using vision–language models (VLMs), depth-grounded 3D projection, cross-view association, and scene retrieval.

## ADDITIONAL RESEARCH AND INDUSTRY EXPERIENCE

**NYU C2SMARTER Center** *Brooklyn, NY*
*Graduate Research Assistant* Sept. 2023 – Mar. 2026

- Developed distributed online optimization and transportation digital-twin methods for predictive camera control and cooperative driver-risk sensing; contributed to papers in IEEE Control Systems Letters, IEEE T-ITS, and *Transportation Research Part C*.

**NYU Center for Cybersecurity** *Brooklyn, NY*
*Graduate Research Assistant* Sept. 2021 – Jan. 2024

- Developed game-theoretic and neuro-symbolic learning for nonstationary autonomous driving, secure route recommendation, and adaptive penetration testing; contributed to papers at IEEE ICRA, IEEE TIFS, and IEEE MILCOM.

**JD.com, Inc.** *Beijing, China*
*Algorithm Engineer Intern* Jan. 2021 – Feb. 2021

- Developed CenterNet/PIoU-based rotated-object detection and supported system integration and on-site validation in a manufacturing laboratory.

## SELECTED PUBLICATIONS AND MANUSCRIPTS

Representative work in localization, robotics, and wireless sensing; complete record on Google Scholar.

### Selected manuscripts

- **MAGNETAR: Multipath-Guided Spatial Posteriors for Transmitter Pose Inference in the Upper Mid-Band.** First and corresponding author; IEEE ICRA 2027, under review. **[paper]**
- **MAPLE-RF: Efficient Probabilistic RF Source Localization in Partially Explored Environments.** First and corresponding author; IEEE ICRA 2027, under review. **[paper]**
- **Posterior Inference for Indoor Localization with Ray-Tracing Wireless Digital Twins.** First and corresponding author; revised and resubmitted to IEEE TWC.
- **Learning a Measurement-to-Posterior Map for Wireless Localization.** First and corresponding author; IEEE TVT, under review.

### Selected peer-reviewed publications

- **LOCUS-DT: Localization via Observation-Conditioned Uncertainty Scoring with Digital Twins.** First and corresponding author; IEEE GLOBECOM 2026, accepted. **[paper]**
- **Zero-Shot Wireless Indoor Navigation through Physics-Informed Reinforcement Learning.** Co-first author; IEEE ICRA 2024. **[paper]**
- **Transformer-Based Rate Prediction for Multi-Band Cellular Handsets.** Co-first and corresponding author; IEEE ICC Workshops 2026. **[paper]**

## SELECTED HONORS

Ernst Weber Fellowship, NYU Tandon School of Engineering (2023) | School of Engineering Fellowship, NYU Tandon (2022) | Tandon Research Excellence Exhibit (2022)